

Castalia

A four-channel electrochemical sensing SoC with a five-hart RISC-V back end

Maxx Seminario

University of Nebraska–Lincoln

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Where cancers are caught, and whether anyone looks

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Target
application

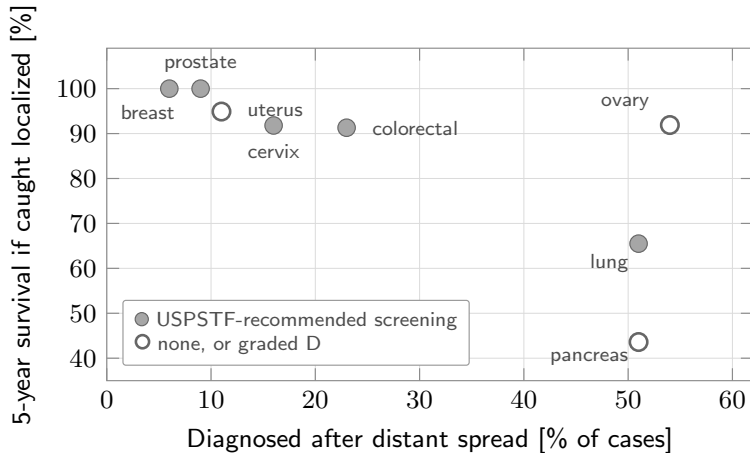
The chip

Analog front end

From currents to
a score

References

Backup



[1] SEER 21, 2016–2022; [2, 3] USPSTF

Why ovarian: late, lethal, and unscreened

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application

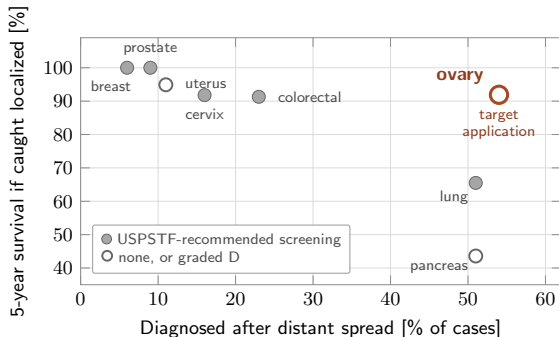
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Site	Distant at dx	Deaths/100 cases	Screening (USPSTF)
Ovary	54 %	59	D
Pancreas	51 %	78	D
Lung	51 %	54	B, LDCT
Colorectal	23 %	35	A
Cervix	16 %	31	A
Uterus	11 %	21	none
Prostate	9 %	11	C, PSA
Breast	6 %	13	B

USPSTF: A/B recommended; C offered selectively; D recommended *against*; none: no statement [7].

[1] SEER, 2026 estimates; [2, 3] USPSTF

Most lethal gynecologic cancer [1]: 92 % survivable if caught localized, yet 54 % present after spread, and screening is graded D [2]. UKCTOCS moved stage, not mortality [4].

One serum aliquot, four sites, one score

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Target application

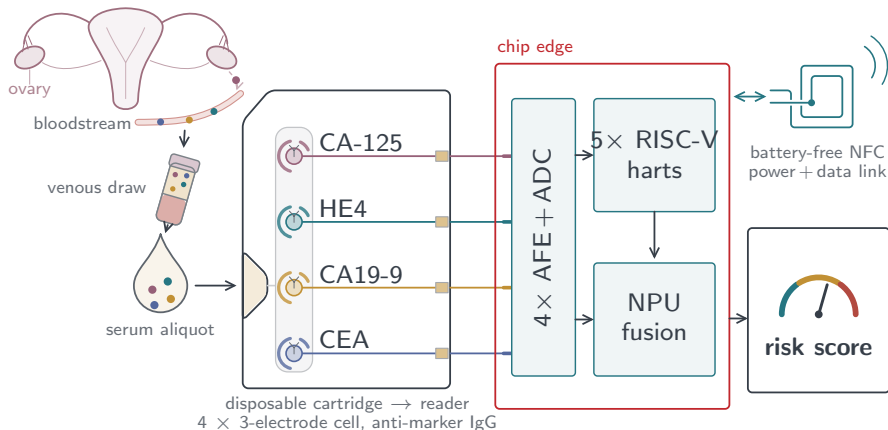
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Symptomatic-patient triage on a disposable four-site cartridge; four markers read at the same instant, fused on chip.

No single ovarian marker is a usable test [5]

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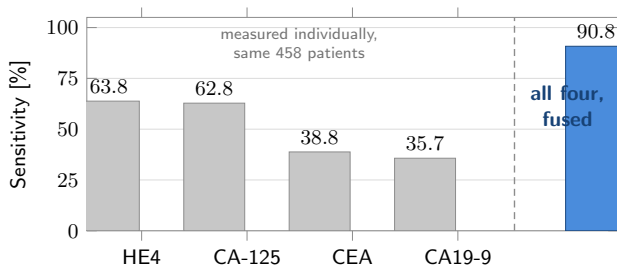
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Panel	Sensitivity
Best single (HE4)	63.8 %
Four fused	90.8 %
ROMA, FDA-cleared [6]	2 markers
This chip	4 channels

Clinical data (458 patients), not simulation. The decision is a learned combination, so the unit of integration is four concurrent channels plus the processor fusing them.

The four markers

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From currents to
scores

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Marker	What it is	Where it fails alone	Cut-off	Sens. [5]
CA-125	epitope on MUC16, a 3–5 MDa membrane mucin shed by ovarian epithelium [8]	raised in only half to two thirds of stage I; confounded by endometriosis, fibroids, ascites [8]	35 U/mL	62.8 %
HE4	human epididymis protein 4 (WFDC2), ~25 kDa secreted glycoprotein; CA-125's partner in ROMA [6]	silent in mucinous tumours	70 pmol/L	63.8 %
CA19-9	sialyl-Lewis ^a carbohydrate antigen carried on mucins; the pancreatic and GI marker	non-specific alone; picks up the mucinous subtype the first two miss	37 U/mL	35.7 %
CEA	carcinoembryonic antigen (CEACAM5), ~180 kDa glycoprotein; the colorectal marker	non-specific alone; mucinous and GI-type tumours	5 ng/mL	38.8 %
All four, fused [5]				90.8 %

Two epithelial markers plus two mucinous/GI markers: the fusion works because each covers a subtype the others miss.

Four markers resolved in one differential-pulse sweep

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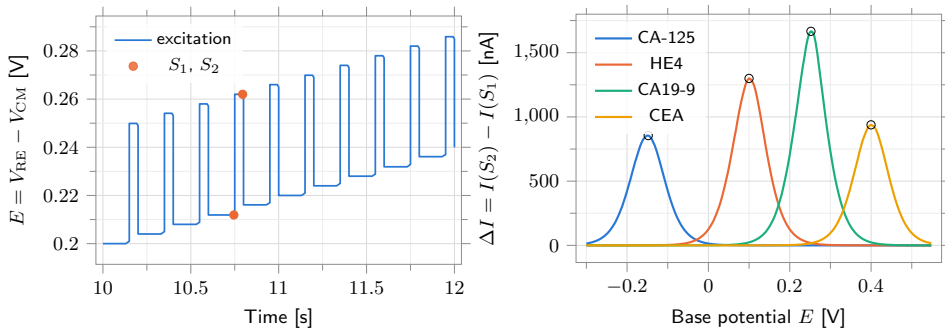
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From currents to a score

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■ schematic sim, pre-layout

Peak position identifies the reporter, peak height carries concentration. Apexes within 2 mV and 1 % of the assay targets; blank below 1 pA.

Every clinical cut-off lands inside the channel window

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Target
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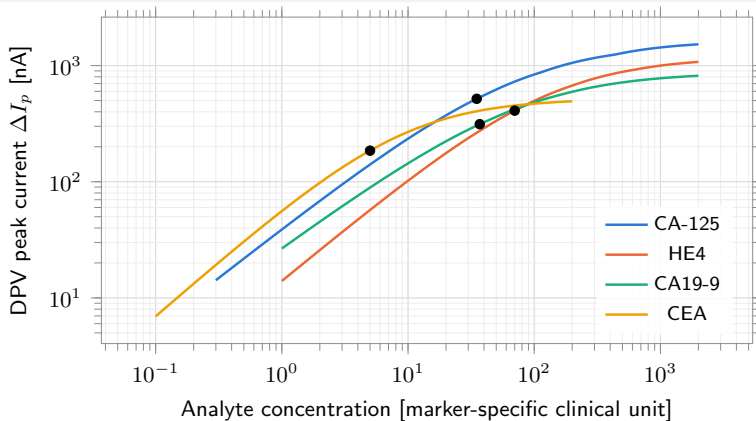
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Analog front end

From currents to
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References

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■ schematic sim,

pre-layout

CA-125 35 U/mL, HE4 70 pmol/L, CA19-9 37 U/mL, CEA 5 ng/mL: all four on the steep part of the curve at one R_f setting.

Castalia at a glance

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Front currents to a scan

References

Backup

Digital

Harts	5 (1 orchestrator, 4 channel tiles)
ISA	rv32imac_zba_zbb_zbs_zbc
Clock	24 MHz rated, closed at 25 MHz
Measured CPI	1.260 aggregate, nine benchmarks
Private TCM	8 KiB per hart, 1-cycle
Shared RAM	80 KiB behind a round-robin arbiter
Synchronisation	16 hardware mutexes, LR/SC, AMOs
Interrupts	claim/complete, any vector to any hart
Accelerator	NPU: MLP, conv-1D, XNOR, GEMM
Power domains	4 MTCMOS, one per channel tile
Debug	JTAG, RISC-V Debug Module
Field boot	NFC ISO 14443A harvested boot

Analog and physical

Measurement sites	4 identical, simultaneous
Topology	bipolar grounded-WE potentiostat
Electrode pads	12 (CE/RE/WE $\times$ 4)
Converter	10-bit SAR per channel, extracted
Reference DAC	12-bit R-2R per channel
Feedback resistor	$18\text{ k}\Omega + 6\text{ k}\Omega \cdot N$, 320 codes
Full scale	$\pm 40.7\text{ }\mu\text{A}$ to $\pm 0.41\text{ }\mu\text{A}$
EIS	firmware mode of the same channels
Analog supply	2.5 V , $v_{\text{cm}} = 1.25\text{ V}$
Process	65 nm CMOS, 1P8M, HVT cells
Die / package	$3.0 \times 3.0\text{ mm}$ / LQFP-100
Digital power	10.72 mW post-route, SS, 25 MHz

Digital numbers from the hardened flow; analog numbers schematic-level, except the SAR, which is extracted.

Block diagram

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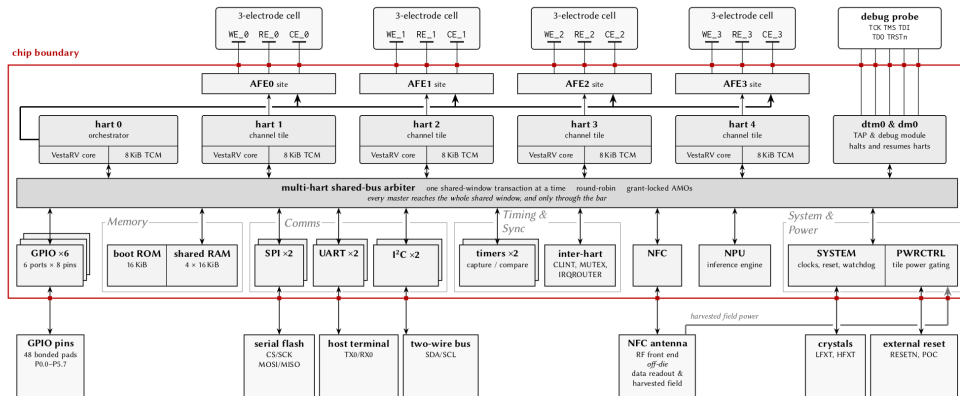
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Analogue front end

From currents to
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Four channel tiles own four AFE sites; every master reaches ROM, RAM, NPU and peripherals only through the arbiter.

Physical implementation

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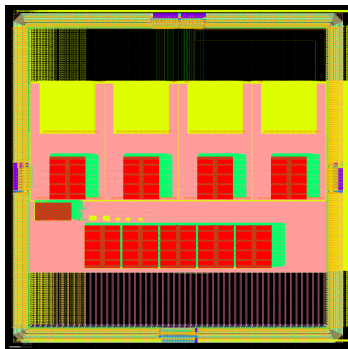
The chip

Analog front end

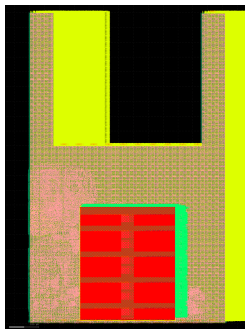
From currents to
a score

References

Backup



chip top, 3.0 × 3.0 mm



one hardened channel tile

Process	65 nm, 1P8M, HVT
Core box	2690 × 2690 μm
Tile	660 × 880 μm
Tile logic	≈98 k NAND2-eq.
Tile SRAM	8 KiB
Package	LQFP-100, 14 × 14 mm

■ hardened flow, post-route

One tile hardened once, instantiated four times; the notch in each tile is the analog channel's keep-out.

Digital, as closed

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Whole chip, post-route

Internal	3.78 mW
Switching	2.33 mW
Leakage	4.61 mW
Total	10.72 mW

One channel tile

Total	1.761 mW
Gated	0

Measured CPI, timed kernels

Benchmark	Instr.	CPI
spmv	802 447	1.143
multiply	20 892	1.163
vvadd	2 408	1.374
median	4 015	1.398
qsort	123 205	1.398
rsort	146 548	1.412
dhrystone	200 523	1.504
towers	3 747	1.682
memcpy	11 017	1.726
Aggregate	1 314 802	1.260

Timing at 25 MHz

reg2reg WNS	+99 ps
Leakage share	43 %

■ hardened flow, post-route

SS 0.9 V 125 °C, statistical activity 0.2. Gating one tile removes 1.76 mW; the always-on band pays nothing.

Grounded-WE two-amplifier potentiostat

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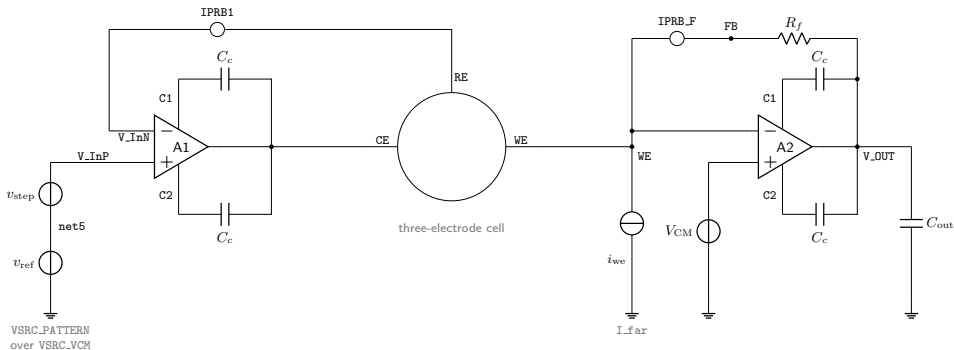
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Analog front end

From currents to
a scope

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A1 drives CE with RE in feedback; A2 holds WE at v_{cm} through R_f . $v_{out} = v_{cm} - I_{we}R_f$, both polarities.

Analog top level

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ARTWORK PENDING



Virtuoso export of the analog top level: four channels, shared bias, test bus

The class-AB amplifier: one cell, both roles

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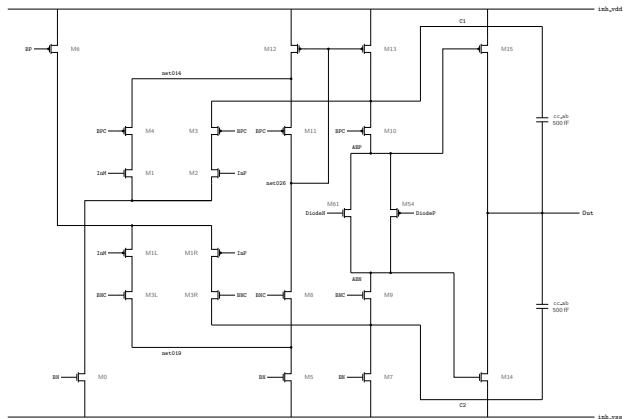
The chip

Analog front end

From currents to
a scope

References

Backup



complementary
input pairs and
their cascodes

diode-connected
summing branch

class-AB
drive

output

testbench
compensation

A1 across corners

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Analog front end

From currents to a score

References

Backup

Parameter	Unit	tt	ff	fs	sf	ss	Spread
Open-loop gain A_{OL}	dB	82.77	79.88	83.14	83.37	84.94	5.06
Phase margin	°	73.52	74.70	74.04	72.86	73.34	1.84
Dominant pole $f_{-3\text{dB}}$	kHz	1.592	2.286	1.538	1.494	1.225	1.061
Gain-bandwidth product	MHz	22.781	23.187	22.894	22.939	22.587	0.600
Parameter	Unit	tt	ff	fs	sf	ss	Spread
Input offset voltage	mV	0.622	0.649	0.624	0.569	0.456	0.193

■ schematic sim, pre-layout

A1 under mismatch

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Target
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Analog front end

From currents to
a score

References

Backup

Parameter	Unit	N	Mean	σ	p_1	p_{99}	Worst	Spec	Yield	4 ch
Open-loop gain A_{OL}	dB	200	82.16	1.06	80.04	84.60	79.50	≥ 40.00	100.0	100.0
Phase margin	$^\circ$	200	73.62	1.42	69.80	76.21	68.93	≥ 40.00	100.0	100.0
Dominant pole $f_{-3\text{dB}}$	kHz	200	1.723	0.276	1.158	2.453	–	–	–	–
Gain-bandwidth product	MHz	200	22.815	3.566	16.191	31.741	–	–	–	–

Parameter	Unit	N	Mean	σ	Mean $\pm 3\sigma$	Min	Max	Spec	Yield	4 ch
Input offset voltage	mV	200	0.583	5.201	0.583 ± 15.604	–14.130	15.709	± 1.000	12.0	0.0

■ Monte Carlo, 200 samples

A2 as transimpedance amplifier: corners and mismatch

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Analog front end

From currents to a score

References

Backup

Parameter	Unit	tt	ff	fs	sf	ss	Spread
Loop gain at DC	dB	78.74	76.32	79.28	79.32	82.69	6.37
Loop crossover	kHz	9.375	9.475	9.428	9.428	9.360	0.115
Phase margin	°	76.50	80.30	76.23	76.13	72.78	7.52
Gain margin	dB	21.81	20.67	21.90	21.24	22.49	1.82

Parameter	Unit	N	Mean	σ	Mean $\pm 3\sigma$	Min	Max	Spec	Yield	4 ch
Output offset at zero current	mV	200	-1.08	8.91	-1.08 $\pm$ 26.72	-28.46	25.28	± 5.00	48.0	5.3

■ schematic sim, pre-layout ■ Monte Carlo, 200 samples

Bipolar accuracy through zero

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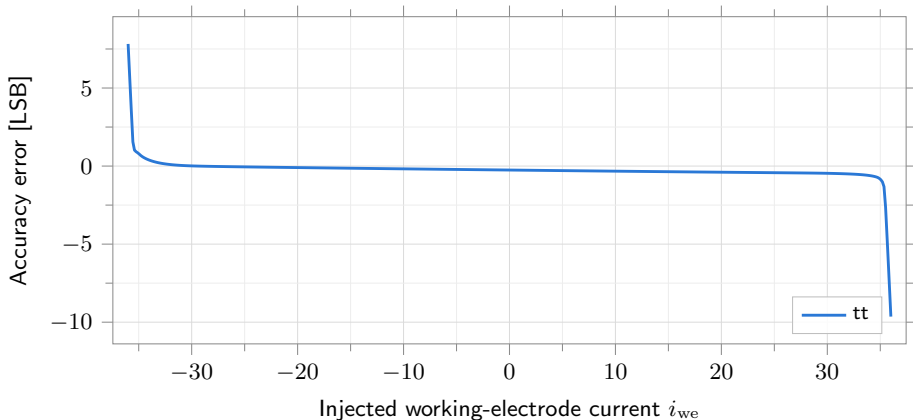
The chip

Analog front end

From currents to a score

References

Backup



■ schematic sim, pre-layout

No kink at the crossing: class-AB output, no dead zone. This is the measurement rev-1's PMOS mirror could not make.

Cyclic voltammetry through the assembled channel

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Target
application

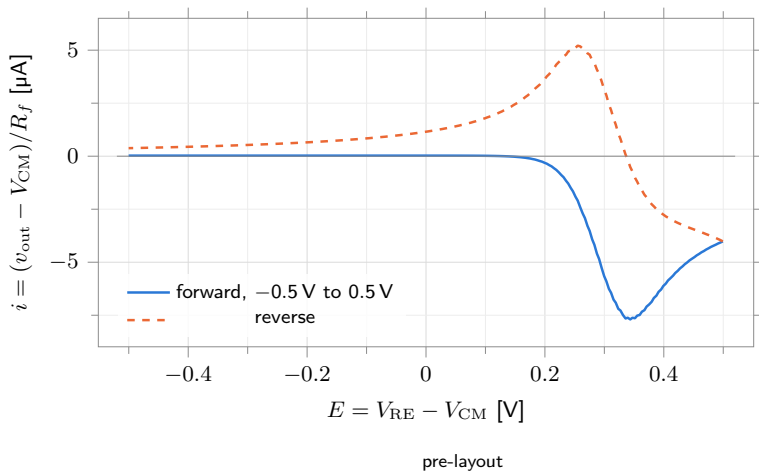
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Analog front end

From currents to
a scope

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Backup



Staircase potential from the on-chip DAC, current digitised by the on-chip SAR, unclipped over the sweep.

Programmable feedback resistor: one string, 320 codes

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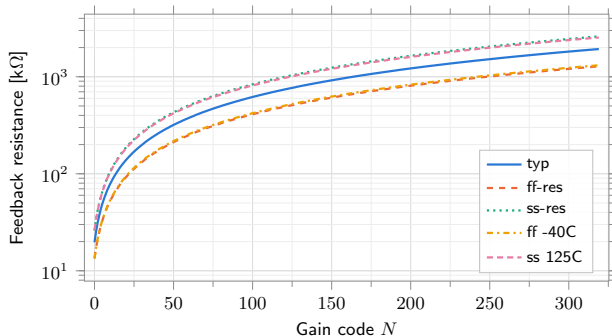
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Analog front end

From currents to a scale

References

Backup



$$R_f = 18 \text{ k}\Omega + 6 \text{ k}\Omega \cdot N$$

N	R_f	Full scale
0	19.6 kΩ	$\pm 40.7 \mu\text{A}$
63	397 kΩ	$\pm 2.02 \mu\text{A}$
319	1.93 MΩ	$\pm 0.41 \mu\text{A}$
LSB at 319		0.824 nA

Strictly monotonic at every corner, including the four thermometer seams. Maximum gain is noise-limited, not quantisation-limited.

Feedback resistor: switch structure, corners, mismatch

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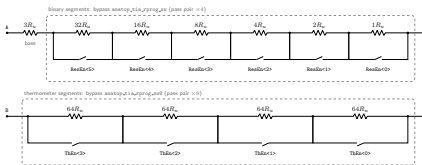
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From currents to a score

References

Backup



Parameter	Unit	typ
Chord resistance	k Ω	19.639
Worst deviation	mV	1.998
Worst deviation	ADC LSB	1.255
Secant- R modulation	%	0.504

Parameter	Unit	N	Mean	σ	Yield
$R(N=0)$	k Ω	200	19.650	0.071	100.0
$R(N=63)$	k Ω	200	396.4515	0.3608	100.0
$R(N=64)$	k Ω	200	403.8759	0.3644	100.0
$R(N=319)$	M Ω	200	1.93212	0.00071	100.0
Seam step 63 $\rightarrow$ 64	k Ω	200	7.424	0.504	100.0

■ schematic sim, pre-layout ■ Monte Carlo, 200 samples

SAR ADC: charge-redistribution architecture

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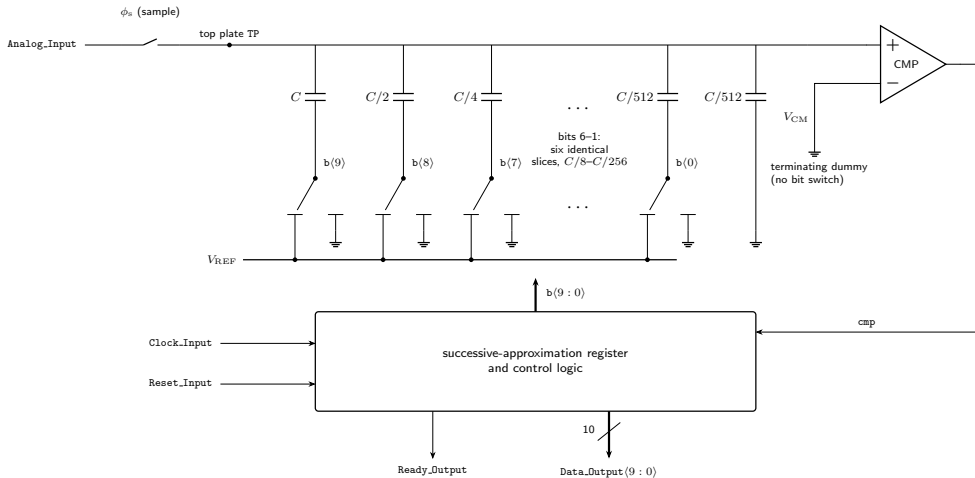
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10-bit SAR ADC, extracted netlist

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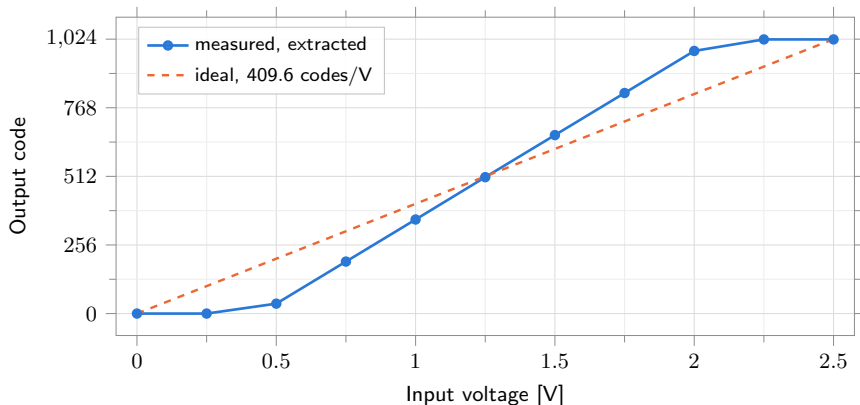
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Analog front end

From currents to
a scope

References

Backup



■ extracted

netlist

23 clocks per conversion, 870 kSa/s, 50 pJ. The capacitor array exists only in layout, so this is the one block characterised extracted.

SAR ADC across corners

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The chip

Analog front end

Front currents to a scope

References

Backup

Parameter	Unit	tt 37 °C	ff	fs	sf	ss
Fitted slope	codes/V	628.59	628.59	628.40	628.73	628.13
Effective LSB	mV	1.5909	1.5909	1.5914	1.5905	1.5920
Input at code 0	V	0.4408	0.4408	0.4407	0.4408	0.4399
Usable span	V	1.6274	1.6274	1.6280	1.6271	1.6286
Worst residual	LSB	0.495	0.495	0.490	0.477	0.484
RMS residual	LSB	0.294	0.294	0.277	0.288	0.302
Monotonic points		20/20	20/20	20/20	20/20	20/20
Conversion period	µs	1.150	1.150	1.150	1.150	1.150
Analog supply current	µA	6.360	4.787	5.453	8.437	15.429
Total power	µW	18.09	16.77	15.77	23.74	43.06
Energy per conversion	pJ	20.8	19.3	18.1	27.3	49.5

■ extracted netlist

Reference DAC: 12-bit voltage-mode R-2R ladder

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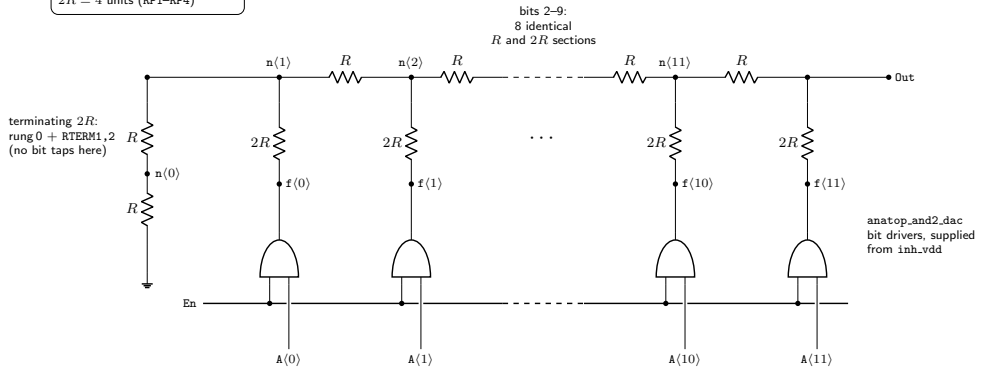
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12 bits, one $2R$ branch each.
Every resistor is one *unit* device:
 $R = 2$ units ($RS1+RS2$),
 $2R = 4$ units ($RP1-RP4$)



Reference DAC across corners and under mismatch

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The chip

Analog front end

Front currents to a scope

References

Backup

Parameter	Unit	tt	ff	fs	sf	ss	Spread
Zero-scale output	μV	0.02	0.25	0.01	0.06	0.00	0.25
Full-scale output	V	2.499388	2.499388	2.499388	2.499388	2.499388	0.000000
LSB size	μV	610.351	610.351	610.351	610.351	610.351	0.000
INL, most positive	LSB	0.113	0.101	0.100	0.127	0.130	0.030
INL, most negative	LSB	-0.376	-0.338	-0.390	-0.362	-0.414	0.076
INL, worst magnitude	LSB	0.376	0.338	0.390	0.362	0.414	0.076
DNL, most positive	LSB	0.002	0.001	0.002	0.002	0.002	0.000
DNL, most negative	LSB	-0.423	-0.374	-0.424	-0.424	-0.479	0.105
DNL, worst magnitude	LSB	0.423	0.374	0.424	0.424	0.479	0.105

Parameter	Unit	N	Mean	σ	Median	p_1	p_{99}	Worst	Spec	Yield	4 ch
INL worst	LSB	200	1.485	0.620	1.364	0.426	3.287	3.393	≤ 1.000	20.5	0.2
INL max	LSB	200	1.154	0.610	1.031	0.194	2.886	-	-	-	-
INL min	LSB	200	-1.485	0.620	-1.364	-3.287	-0.426	-	-	-	-
DNL min	LSB	200	-1.682	1.415	-1.287	-6.161	-0.058	-6.499	≥ -1.000	42.0	3.1
DNL max	LSB	200	1.123	1.128	0.713	0.007	4.760	-	-	-	-
DNL worst	LSB	200	2.249	1.314	1.847	0.441	6.161	6.499	≤ 0.750	9.0	0.0

■ schematic sim, pre-layout ■ Monte Carlo, 200 samples

Local bias generator

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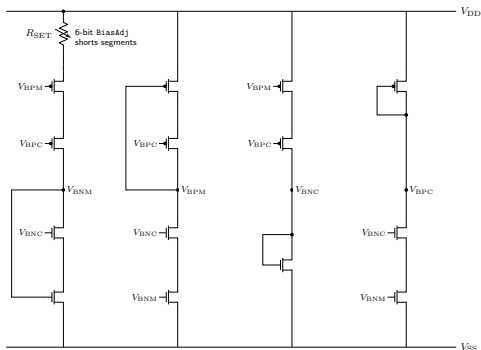
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From currents to a score

References

Backup



■ schematic sim, pre-layout

Parameter	Unit	Min	Mean	Max	Spread
V_{BNM}	mV	569.9	593.8	618.1	48.2
V_{BNC}	mV	766.3	771.5	777.1	10.8
V_{BPM}	mV	1811.2	1838.4	1865.4	54.2
V_{BPC}	mV	1589.2	1590.0	1590.9	1.7
I_{DD}	μA	24.16	25.50	26.81	2.65

Impedance spectroscopy on the same channel

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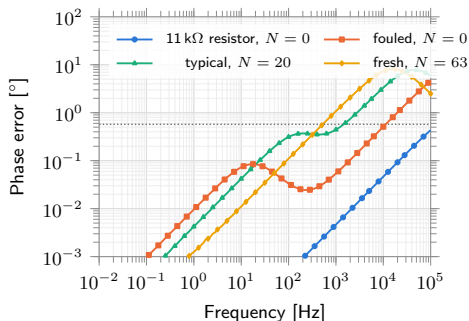
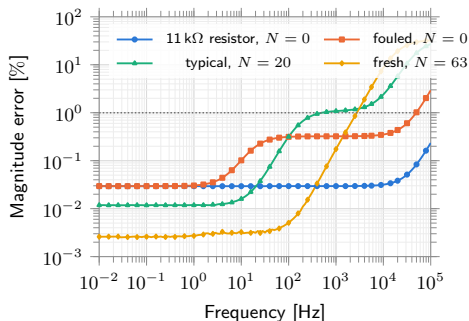
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From currents to a scope

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Excitation	Demodulation	Accuracy	$ Z $ range
2-code DAC square	firmware, channel hart	$\pm 1\%$, $\pm 0.6^\circ$ to 5.05 kHz; $\pm 5\%$ to 26.5 kHz	246 Ω – 12.1 M Ω

■ schematic sim, pre-layout

No EIS engine. Impedance is a ratio of two readings through one converter; the DAC drops out of the accuracy budget.

One AFE channel, as simulated

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Parameter	Value	Condition
CE compliance	13.2 mV – 2.480 V	2.5 V rail, corners
Cell current delivered	$\pm 52.5 \mu\text{A}$	12 k Ω cell
TIA compliance	$\pm 0.80 \text{ V}$ about v_{cm}	
Feedback resistor	19.6 k Ω – 1.93 M Ω , 320 codes	monotonic at every corner
Full-scale current	$\pm 40.7 \mu\text{A}$ ($N=0$) to $\pm 0.41 \mu\text{A}$ ($N=319$)	SAR window $\cap$ TIA compliance
Current quantum	0.824 nA at maximum gain	below the 2 nA noise budget
Bipolar accuracy	0.25–0.38 LSB worst	560 k Ω , 1 M Ω electrode, corners
Control-loop PM	79–80°	five corners, into a cell
TIA-loop PM	89–93°	12-cell electrode grid
Input current noise	341 pA standalone, 3.6–21 nA into an electrode	0.1–100 Hz, 560 k Ω
Zero-current offset	$\sigma(v_{\text{re}}) = 5.72 \text{ mV}$, gain-independent	200 MC samples, calibratable
Quiescent current	197–250 μA per channel	amplifiers + local bias

■ schematic sim, pre-layout ■ Monte Carlo, 200 samples

Pre-layout, electrode held nominal: no row includes electrode-to-electrode spread.

The fusion runs on the NPU

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$$s = \sigma \left(\sum_{k=1}^4 w_k \hat{c}_k + b \right)$$

Input	peak current, peak potential, baseline $\times 4$
Mode	0, dense layer, one sigmoid
Output	one risk score
Weights	trained off-chip, staged by firmware

Per site

Hart	1, ownership enforced in hardware
Potential	own 12-bit DAC
Converter	own 10-bit SAR
Power	own MTCMOS domain

The logistic form of the cleared two-marker algorithm, extended to four inputs and moved from the central laboratory onto the electrode.

References

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References

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Backup: USPSTF screening grades

Castalia

Maxx Seminario

Target application

The chip

Analog front end

From currents to a score

References

Backup

Grade	Certainty of net benefit	Definition [7]	Cancer screening [2, 3]
A	high, substantial	Recommends the service. Offer or provide it.	cervix (2018), colorectal 50–75 (2021)
B	high moderate, or moderate	Recommends the service. Offer or provide it.	breast 40–74 (2024), lung LDCT 50–80 (2021), colorectal 45–49 (2021)
C	at least moderate, small	Offer selectively, by professional judgment and patient preference.	prostate PSA 55–69 (2018)
D	no net benefit, or harms outweigh	Recommends <i>against</i> the service. Discourage its use.	ovary (2018) , pancreas (2019)
I	cannot be determined	Evidence insufficient, poor, or conflicting.	—
none	—	No USPSTF statement exists for the condition.	uterus

Ovarian D is a verdict on a test that was tried, CA-125 plus transvaginal ultrasound [2, 4], not on the absence of one.